# TUBERCULOSIS OF THE BONES

# AETIOLOGY

- It is caused by Mycobacterium tuberculosis

**ROUTES OF INFECTION( MODES OF TRANSMISSION)**

1. Blood stream(haematogenous)from pulmonary foci
2. Direct extension from a joint or soft tissue
3. Rarely- lymphatic spread from infected lymphnodes

# PATHOLOGY

- When the bacilli reaches the bone, there a typical tuberculous inflammatory reaction.
- Part of the bone is destroyed and replaced by granulation tissue.
- A tuberculous abscess is commonly formed. It tracks beneath the soft tissues or towards the surface of the body.
- With treatment, there is a tendency to healing by fibrosis.

# SITES

## 1) Vertebral column( thoracic & lumbar vertebrae)

- It’s the commonest site
- The infection typically affects the vertebral bodies
- The vertebral bodies collapse anteriorly becoming wedge shaped resulting to **Gibbus deformity**( exaggerated kyphosis)
- An abscess may track downwards along the vertebral column
- The abscess may extend backward towards the spinal canal causing pressure on the spinal cord which may lead to paraplegia

# Sites cont'd

**2) Hands**

- It affects the metacarpals and phalanges causing tuberculous dactylitis

**3) Feet**

- It affects the metatarsals and phalanges

**4) Long bones( femur, tibia, radius & ulna)**

- It causes **juxta-articular** tuberculosis
- It occurs as an extension of tuberculous arthritis
- The articular ends of the bones are frequently eroded by tuberculosis.

# CLINICAL FEATURES

It has a gradual( insidious) onset.

**1. Constitutional features:**

- Night sweats
- Fever
- General malaise
- Loss of weight
- Some may have a cough(rare)

# Clinical features cont'd

## 2. Specific features:

The depend on the site affected:

- Hands and feet- Pain and swelling
- Long bones- pain and stiffness of affected joint
- Vertebrae- Backache, deformities( scoliosis or kyphotic Gibbus deformity)

- Cold abscess as the pus follows the psoas muscle

- Paraplegia and other neurological features

# INVESTIGATIONS

1) **Full haemogram + ESR**

- Hb- slightly reduced
- Relative lymphocytosis
- ESR- moderately elevated( 40-60mm/hr)

2) **Sputum** for AAFB

3) **Gastric lavage** for AAFB

# Investigations cont'd

## 4) Radiological investigations:

**a. X-ray**

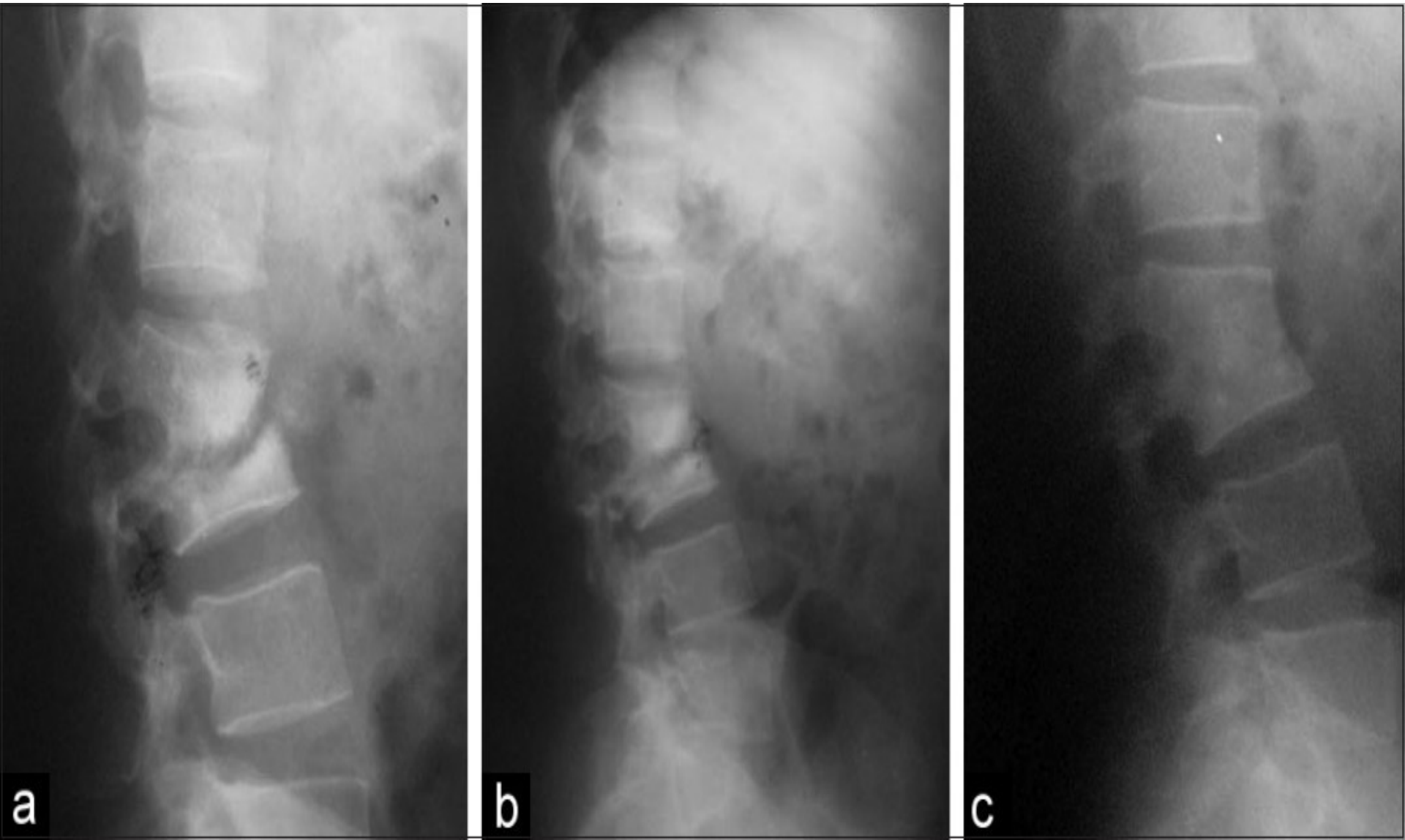
- Diffuse rarefaction around the site of infection
- Erosion of the bone(juxta-articular tuberculosis)
- In most cases – shadow in soft tissues denoting abscess formation.
- Wedging of the vertebral bodies

**b) MRI-** Bone destruction, abscess, spinal cord involvement

**c) CT Scan**

**d) Radioisotope scanning-** areas or increased osteoblastic activity( red spots)

# Tb spine with wedging of vertebrae



# Tb of acetabulum & femoral head

# Tb of the metatarsals & phalanges

# Investigations cont'd

**5) Mantoux test-** positive

**6) Pus aspirate** for culture

**7) Biopsy of the bone or soft tissues for histology** – features of tuberculosis

# TREATMENT

## 1) Supportive treatment

- Nutritional support
- Correct the anaemia
- Analgesics

## 2) Chemotherapy

- Combined anti-Tb therapy for six months. Drugs are combined so a to:
  - reduce resistance
  - reduce the dosage of individual drugs
  - reduce drug complications

# Treatment cont'd

## 3) Surgery

- Drainage of the abscess
- Surgical decompression of the spinal cord and stabilization of vertebral column
- Bone curettage and grafting
- Synovectomy
- Arthroplasties

# Treatment cont'd

## 4) Rehabilitation

- Physiotherapy
- Use of crutches
- Provision of wheelchairs for paraplegics

# PROGNOSIS

The outcome of the disease may take three forms:

i. Heal completely
ii. Become a chronic illness
iii. Spread to other organs in form of milliary Tb

# COMPLICATIONS

a) Permanent bone/joint destruction leading to permanent deformities
b) Paraplegia and other neurological complications
c) Spread to other organ- milliary Tb

# ASSIGNMENTS

1. Syphilitic bone infection
2. Madurella mycetoma ( Madura foot)

# QUOTE!!

"SO MANY PEOPLE SPEND THEIR HEALTH GAINING WEALTH, AND THEN HAVE TO SPEND THEIR WEALTH TO REGAIN THEIR HEALTH"

A.J. Reb Materi

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